SIEIS Project Overview

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What it is

SIEIS is an ESP32 that reads indoor sensors and sends the readings over Wi-Fi to a laptop. A Python backend stores them in SQLite, and a web dashboard shows live values, history graphs, trends and alerts.

What it measures

Only temperature and humidity are real physical values; the other three are relative signals and must not be labelled as calibrated units.

Sensor	Reading	Treat it as	Not
DHT22	Temperature, humidity	Real values, reasonable accuracy	Pressure (the BME280 was dropped)
MQ-135	Gas response	Relative change from its own baseline	ppm, CO_2 or AQI
Bare LDR	Brightness	Dark / dim / bright and daily trends	Lux
Sound module	Noise activity	Quiet versus clap or talking	Decibels

Where the build stands

The ESP32 and the sound module are confirmed working, but the combined sketch with all four sensors is not yet confirmed.

- **Verified:** ESP32 with the CP210x USB driver on COM4, the sound module's AO output and threshold, and the original light module's DO output.
- **Not yet confirmed:** the combined sketch reading the DHT22, LDR, MQ-135 and sound module together. No output from it has been checked.
- **MQ-135:** it still needs a 24 to 48 hour burn-in before its readings are stable.

What's left, in order

The next step is checking all sensors together; each later step builds on the one before it.

1. **All sensors together:** run the combined sketch and confirm every value responds correctly in the Serial Monitor.
2. **Wi-Fi:** connect the ESP32 to the network and check it reaches the laptop.
3. **JSON over HTTP:** the ESP32 posts readings every 5 to 10 seconds.
4. **Python backend:** a small Flask or FastAPI server receives the JSON and writes it to SQLite.
5. **Dashboard:** live value cards plus historical graphs.
6. **Analysis and alerts:** moving averages, trends, a temperature and humidity comfort rating, and an MQ-135 "worse than baseline" alert.

Decisions to make early

Two choices affect the backend and the dashboard, so settle them before building those.

- **Pressure:** drop it from the project, or add a BMP280 or BME280 with soldered header pins later. Adding one later would not change the rest of the design.
- **MQ-135 baseline:** alerts compare against the sensor's own rolling average, so the backend needs to compute and store that average.